

FACTORIZATION, SIMULTANEOUS EQUATIONS

6.1 BASIC ALGEBRAIC FORMULAS

6.1.1 Recall of the formulas:

$$(i) \quad (a + b)^2 = a^2 + 2ab + b^2$$

$$(ii) \quad (a - b)^2 = a^2 - 2ab + b^2$$

$$(iii) \quad a^2 - b^2 = (a - b)(a + b)$$

We have already studied and verified the above mentioned formulas in the previous class.

Let us apply them to solve some problems:

Example 1. Evaluate: (i) $(102)^2$ (ii) $(98)^2$ (iii) $(102)(98)$

Solution:

$$\begin{aligned} (i) \quad (102)^2 &= (100 + 2)^2 \\ &= (100)^2 + 2(100)(2) + (2)^2 \\ &\quad [\text{using } (a + b)^2 = a^2 + 2ab + b^2] \\ &= 10000 + 400 + 4 = 10404 \end{aligned}$$

$$\begin{aligned} (iii) \quad (102)(98) &= (100 + 2)(100 - 2) \\ &= (100)^2 - (2)^2, \end{aligned}$$

$$\begin{aligned} \text{Using: } [(a + b)(a - b) &= a^2 - b^2] \\ &= 10000 - 4 = 9996 \end{aligned}$$

$$\begin{aligned} (ii) \quad (98)^2 &= (100 - 2)^2 \\ &= (100)^2 - 2(100)(2) + (2)^2, \\ &\quad [\text{using } (a - b)^2 = a^2 - 2ab + b^2] \\ &= 10000 - 400 + 4 = 9604 \end{aligned}$$

Activity: Solve 203×197

$$\begin{aligned} &(200 + \text{---}) \times (\text{---} - 3) \\ &\text{using } (a + b)(a - b) = (\text{---}) \\ &= (200)^2 - (\text{---})^2 \text{---} \\ &= 40,000 - \text{---} = \text{---} \end{aligned}$$

Example 2. Evaluate: (i) $(1.02)^2$ (ii) $(0.98)^2$ (iii) $(1.02)(0.98)$

Solution:

$$\begin{aligned} (i) \quad (1.02)^2 &= (1 + .02)^2 \\ &= (1)^2 + 2(1)(.02) + (.02)^2, \\ \text{[Using: } \therefore (a + b)^2 &= a^2 + 2ab + b^2] \\ &= 1 + 0.04 + 0.0004 = 1.0404 \end{aligned}$$

$$\begin{aligned} (ii) \quad (0.98)^2 &= (1 - 0.02)^2 \\ &= (1)^2 - 2(1)(0.02) + (0.02)^2, \\ \text{[using } (a - b)^2 &= a^2 - 2ab + b^2] \\ &= 1 - 0.04 + 0.0004 = 0.9604 \end{aligned}$$

$$(iii) \quad (1.02)(0.98) = (1 + 0.02)(1 - 0.02) = (1)^2 - (0.02)^2,$$

$$\begin{aligned} \text{[Using: } (a + b)(a - b) &= a^2 - b^2] \\ &= 1 - 0.0004 = 0.9996 \end{aligned}$$

Example 3. Find the value $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$ when (i) $x + \frac{1}{x} = 6$
(ii) $x - \frac{1}{x} = 2$

Solution:

(i) $x + \frac{1}{x} = 6$... Given (Squaring both sides)
 $\Rightarrow (x + \frac{1}{x})^2 = (6)^2$
 $\Rightarrow x^2 + 2(x)(\frac{1}{x}) + \frac{1}{x^2} = 36$
 $\Rightarrow x^2 + 2 + \frac{1}{x^2} = 36$
 $\Rightarrow x^2 + \frac{1}{x^2} = 36 - 2$
 $\Rightarrow x^2 + \frac{1}{x^2} = 34$ (Squaring both sides)
 $\Rightarrow (x^2 + \frac{1}{x^2})^2 = (34)^2$
 $\Rightarrow x^4 + 2(x^2)(\frac{1}{x^2}) + \frac{1}{x^4} = 1156$
 $\Rightarrow x^4 + \frac{1}{x^4} = 1156 - 2 = 1154$

(ii) $x - \frac{1}{x} = 2$... given (Squaring both sides)
 $\Rightarrow (x - \frac{1}{x})^2 = (2)^2$
 $\Rightarrow x^2 - 2(x)(\frac{1}{x}) + \frac{1}{x^2} = 4$
 $\Rightarrow x^2 - 2 + \frac{1}{x^2} = 4$
 $\Rightarrow x^2 + \frac{1}{x^2} = 4 + 2$
 $\Rightarrow x^2 + \frac{1}{x^2} = 6$ (Squaring both sides)
 $\Rightarrow (x^2 + \frac{1}{x^2})^2 = (6)^2$
 $\Rightarrow x^4 + 2(x^2)(\frac{1}{x^2}) + \frac{1}{x^4} = 36$
 $\Rightarrow x^4 + \frac{1}{x^4} = 36 - 2 = 34$

Example 4.

Show that $x^4 + \frac{1}{x^4} = (x^2 + \frac{1}{x^2})^2 - 2$

Proof: R.H.S = $(x^2 + \frac{1}{x^2})^2 - 2$

$$= (x^2)^2 + 2(x^2)(\frac{1}{x^2}) + (\frac{1}{x^2})^2 - 2$$

$$= x^4 + 2 + \frac{1}{x^4} - 2 = x^4 + \frac{1}{x^4} = \text{L.H.S. Hence proved.}$$

EXERCISE 6.1

1. Evaluate the following by using suitable formula:

(i) $(105)^2$

(ii) $(96)^2$

(iii) $(57)^2$

(iv) $(52)^2$

(v) 104×96

(vi) 47×53

(vii) 107×93

(viii) 89×111

2. Evaluate the following by using suitable formula:

- (i) $(1.03)^2$ (ii) $(0.99)^2$ (iii) $(1.05)^2$
 (iv) $(0.91)^2$ (v) 1.03×0.97 (vi) 5.02×4.98

3. Find the value of $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$ when:

- (i) $x + \frac{1}{x} = 5$ (ii) $x + \frac{1}{x} = -4$ (iii) $x + \frac{1}{x} = 7$

4. Find the value of $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$ when:

- (i) $x - \frac{1}{x} = 3$ (ii) $x - \frac{1}{x} = 0.2$ (iii) $x - \frac{1}{x} = -6$

5. Show that: (i) $x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2$ (ii) $x^2 + \frac{1}{x^2} = (x - \frac{1}{x})^2 + 2$

6.2 FACTORIZATION

If the product of two or more expressions is equal to the given expression, then these two or more expressions are called the factors of the given expression.

For example; if $3x^2 + 6x = 3x(x + 2)$, then 3, x and $(x + 2)$ are the factors of $3x^2 + 6x$.

The process of finding factors of a given expression is called **factorization**. We express the given expression as a product of two or more expressions.

6.2.1 Factors of the expression of the type: $ka + kb + kc$

We know that if k is a non-zero real number then

$$ka + kb + kc = k(a + b + c)$$

Here k and $(a + b + c)$ are the factors of $k(a + b + c)$

Example 1. Factorize: (i) $5x + 10y + 20z$ (ii) $6x^2 + 12xy - 30xy^2$

Solution: (i) $5x + 10y + 20z$

$$= 5(x + 2y + 4z),$$

($\because 5$ is a common factor)

(ii) $6x^2 + 12xy - 30xy^2$

$$= 6x(x + 2y - 5y^2),$$

($\because 6x$ is a common factor)

6.2.2 Factors of the expression of the type: $ac + ad + bc + bd$

We show that:

$$ac + ad + bc + bd = (a + b)(c + d)$$

Proof: L.H.S. = $ac + ad + bc + bd = a(c + d) + b(c + d)$
 $= (a + b)(c + d) = \text{R.H.S.}$

Example 2. Factorize: (i) $5x + xz + 5z + z^2$ (ii) $3x^2y + 6xy^2 - 2xz - 4yz$

Solution: (i) $5x + xz + 5z + z^2$
 $= x(5 + z) + z(5 + z)$
 $= (5 + z)(x + z)$

(ii) $3x^2y + 6xy^2 - 2xz - 4yz$
 $= 3xy(x + 2y) - 2z(x + 2y)$
 $= (x + 2y)(3xy - 2z)$

6.2.3 Factors of the expression of the type: $a^2 + 2ab + b^2$

(i) We show that:

$$a^2 + 2ab + b^2 = (a + b)^2$$

Proof: L.H.S. = $a^2 + 2ab + b^2$
 $= a^2 + ab + ab + b^2$
 $= a(a + b) + b(a + b)$
 $= (a + b)(a + b)$
 $= (a + b)^2 = \text{R.H.S.}$

(ii) We also show that:

$$a^2 - 2ab + b^2 = (a - b)^2$$

Proof: L.H.S.
 $= a^2 - 2ab + b^2$
 $= a^2 - ab - ab + b^2$
 $= a(a - b) - b(a - b)$
 $= (a - b)(a - b) = (a - b)^2 = \text{R.H.S.}$

Example 1. Factorize: $x^2 + 6xy + 9y^2$

Solution: $x^2 + 6xy + 9y^2$
 $= (x)^2 + 2(x)(3y) + (3y)^2$
 $= (x + 3y)^2, \text{ (using formula)}$

Example 2. Factorize:

$$25x^2 - 10xy + y^2$$

Solution: $25x^2 - 10xy + y^2$
 $= (5x)^2 - 2(5x)(y) + (y)^2 = (5x - y)^2$

6.2.4 Factors of the expression of the type: $a^2 - b^2$

We show that:

$$a^2 - b^2 = (a + b)(a - b)$$

Proof: R.H.S. = $(a + b)(a - b)$
 $= a(a - b) + b(a - b)$
 $= a^2 - ab + ab - b^2$
 $= a^2 - b^2 = \text{L.H.S.}$

Example 1.Factorize: $4a^2 - 9b^2$

Solution: $4a^2 - 9b^2$
 $= (2a)^2 - (3b)^2$
 $= (2a + 3b)(2a - 3b)$

Example 2. Factorize: (i) $25x^2 - 36y^2$ (ii) $8a^2 - 50b^2$

Solution: (i) $25x^2 - 36y^2$

$$= (5x)^2 - (6y)^2$$

$$= (5x + 6y)(5x - 6y)$$

(ii) $8a^2 - 50b^2$

$$= 2(4a^2 - 25b^2)$$

$$= 2\{(2a)^2 - (5b)^2\}$$

$$= 2(2a + 5b)(2a - 5b)$$

6.2.5 Factors of the expression of the type: $a^2 \pm 2ab + b^2 - c^2$

In this type of expressions, we first apply the formula

$$x^2 + 2xy + y^2 = (x + y)^2 \text{ and then } a^2 - b^2 = (a + b)(a - b)$$

Example 1. Factorize $x^2 + 2(x)(5) + 25 - y^2$

Solution: $x^2 + 10x + 25 - y^2 = [(x)^2 + (x)(10) + (5)^2] - y^2$

$$= (x + 5)^2 - y^2 = (x + 5 + y)(x + 5 - y)$$

Similarly we factorize the expression of the type $a^2 - 2ab + b^2 - c^2$ by using the formulas $a^2 - 2ab + b^2 = (a - b)^2$ and $x^2 - y^2 = (x + y)(x - y)$

Example 2. Find factors of $x^2 - 2xy + y^2 - z^2$

Solution: $x^2 - 2xy + y^2 - z^2 = [(x)^2 - 2(x)(y) + (y)^2] - z^2$

$$= (x - y)^2 - z^2 = (x - y + z)(x - y - z)$$

EXERCISE 6.2

A. Factorize the following:

(1) $4x + 8z$

(3) $2x - 4xy + 8xz$

(5) $3a^2b + 7ab^2 - 8a^2b^2$

(7) $5x + 10y + 3xz + 6yz$

(9) $x^2 + 5x + 6xy + 30y$

(11) $3x^2 + 6y^2 + 6x^2z + 12y^2z$

(2) $5x + 10y + 30z$

(4) $2a^2 + 10a^3 - 20a^4$

(6) $6a^2bc + 12ab^2c - 36abc^2$

(8) $abc - abd + cx - xd$

(10) $7xy + 14yz - 5ax - 10az$

(12) $x^2 - 7xy - xz + 7yz$

B. Resolve into factors:

(1) $a^2 + 10a + 25$

(2) $x^2 + 12xy + 36y^2$

(3) $4x^2 + 12xy + 9y^2$

(4) $16a^2 + 40ab + 25b^2$

(5) $b^2 + c^2 - 2bc$

(6) $49p^2 - 14p + 1$

(7) $81c^2 - 36cd + 4d^2$

(8) $144x^4 - 72x^2y^2 + 9y^4$

(9) $3a^2 - 6ab + 3b^2$

(10) $x^2 - 3xy + y^2$

C. Find factors of the following:

(1) $81x^2 - 4y^2$

(2) $169a^2 - 100b^2$

(3) $3a^2 - 27b^4$

(4) $2p^2 - 18q^2$

(5) $5x^2 - 125y^2$

(6) $\frac{25}{x^2} - \frac{y^2}{16}$

(7) $\frac{x^2}{144} - y^2$

(8) $\frac{36}{25} l^2 - \frac{49}{4} d^2$

(9) $c^2 - (a-b)^2$

(10) $(x+y)^2 - z^2$

(11) $(a+b)^2 - (p+q)^2$

(12) $144 - y^4$

(13) $32a^2 - 50b^2$

(14) $x^4 - y^4$

(15) $(2a+b) - 9c^2$

D. Resolve into factors:

(1) $x^2 + 2xy + y^2 - z^2$

(2) $a^2 + b^2 - 2ab - c^2$

(3) $64a^2 + 48ab + 9b^2 - c^2$

(4) $4p^2 - 12pq + 9q^2 - 49r^2$

(5) $x^4 + 2x^2y^2 + y^4 - 25z^2$

(6) $\frac{x^2}{4} - xy + y^2 - \frac{c^2}{36}$

(7) $2a^2 + 4ab + 2b^2 - 2c^2$

(8) $81c^2 - 16p^2 + 18cd + d^2$

(9) $4x^2 + 12x + 9 - 16y^2$

(10) $36 - 25a^2 + 70ab - 49b^2$

6.3 MANIPULATION OF ALGEBRAIC EXPRESSION

Now we recognize and apply following formulas to solve problems.

6.3.1 Formula for the cube of the sum of two terms

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Proof: L.H.S. $= (a+b)^3 = (a+b)(a+b)^2 = (a+b)(a^2 + 2ab + b^2)$
 $= a(a^2 + 2ab + b^2) + b(a^2 + 2ab + b^2)$
 $= a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3$
 $= a^3 + 3a^2b + 3ab^2 + b^3 = \text{R.H.S.}$

Note: $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

$\Rightarrow a^3 + b^3 = (a+b)^3 - 3ab(a+b)$

Example 1. Find the cube of $(2x + 3)$

Solution: $(2x + 3)^3 = (2x)^3 + 3(2x)^2(3) + 3(2x)(3)^2 + (3)^3$
 $= 8x^3 + 3(4x^2)(3) + 6x(9) + 27$
 $= 8x^3 + 36x^2 + 54x + 27$

Example 2. Find the value of $x^3 + \frac{1}{x^3}$ when $x + \frac{1}{x} = 5$

Solution: We have $x + \frac{1}{x} = 5$

Method I: From known to unknown.

Cubing both sides, by formula

$$\left(x + \frac{1}{x}\right)^3 = (5)^3$$

$$x^3 + 3(x^2)\left(\frac{1}{x}\right) + 3(x)\left(\frac{1}{x^2}\right) + \frac{1}{x^3} = 125$$

$$x^3 + 3x + \frac{3}{x} + \frac{1}{x^3} = 125$$

$$\Rightarrow x^3 + 3\left(x + \frac{1}{x}\right) + \frac{1}{x^3} = 125$$

$$x^3 + 3(5) + \frac{1}{x^3} = 125 \quad \left(\because x + \frac{1}{x} = 5\right)$$

$$\Rightarrow x^3 + \frac{1}{x^3} = 125 - 15 = 110.$$

Method II:

From unknown to known.

We know that:

$$a^3 + b^3 = (a + b)^3 - 3ab(a + b)$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x\right)\left(\frac{1}{x}\right)\left(x + \frac{1}{x}\right)$$

$$\Rightarrow x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$$

$$= 5^3 - 3(5)$$

$$= 125 - 15$$

$$= 110.$$

6.3.2 Formula for the cube of the difference of two terms

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Proof: L.H.S. $= (a - b)^3 = (a - b)(a - b)^2 = (a - b)(a^2 - 2ab + b^2)$

$$= a(a^2 - 2ab + b^2) - b(a^2 - 2ab + b^2) = a^3 - 2a^2b + ab^2 - a^2b + 2ab^2 - b^3$$

$$= a^3 - 3a^2b + 3ab^2 - b^3 = \text{R.H.S.}$$

Example 1. Find the cube of $(4x - 1)$

Solution: $(4x - 1)^3$
 $= (4x)^3 - 3(4x)^2(1) + 3(4x)(1)^2 - (1)^3$
 $= 64x^3 - 3(16x^2) + 12x - 1$
 $= 64x^3 - 48x^2 + 12x - 1$

Example 2. Find the cube of 19 using the formula.

Solution: $(19)^3 = (20 - 1)^3$
 $= (20)^3 - 3(20)^2(1) + 3(20)(1)^2 - (1)^3$
 $= 8000 - 1200 + 60 - 1 = 6,859$

6.3.3 Find $x^3 \pm \frac{1}{x^3}$ when the value of $x \pm \frac{1}{x}$ is given

Example 3. Find the value of $x^3 - \frac{1}{x^3}$ when $x - \frac{1}{x} = 2$

Solution: $x - \frac{1}{x} = 2$

Method I: Known to unknown.

$$\begin{aligned} (x - \frac{1}{x})^3 &= (2)^3 && \text{Cubing both sides,} \\ \Rightarrow x^3 - 3(x^2)(\frac{1}{x}) + 3(x)(\frac{1}{x^2}) - \frac{1}{x^3} &= 8, \\ \Rightarrow x^3 - 3x + \frac{3}{x} - \frac{1}{x^3} &= 8 \\ \Rightarrow x^3 - 3(x - \frac{1}{x}) - \frac{1}{x^3} &= 8 \\ \Rightarrow x^3 - 3(2) - \frac{1}{x^3} &= 8 \quad (\because x - \frac{1}{x} = 2) \\ \Rightarrow x^3 - 6 - \frac{1}{x^3} &= 8 \quad \text{or} \quad x^3 - \frac{1}{x^3} = 8 + 6 = 14 \end{aligned}$$

Method II:

Unknown in terms of known.

$$\begin{aligned} a^3 - b^3 &= (a - b)^3 + 3ab(a - b) \\ x^3 - \frac{1}{x^3} &= (x - \frac{1}{x})^3 + 3x \frac{1}{x} (x - \frac{1}{x}) \\ &= (x - \frac{1}{x})^3 + 3(x - \frac{1}{x}) \\ &= (2)^3 + 3(2) = 8 + 6 = 14 \end{aligned}$$

EXERCISE 6.3

1. Find the cube of each of the following:

- (i) $2x + 3$ (ii) $5x - 1$ (iii) $2x - 3y$ (iv) $2a + 5b$ (v) $x + 7z$
 (vi) $\frac{x}{2} - 1$ (vii) $5 + 3a$ (viii) $5 - \frac{a}{4}$ (ix) $5a - 3b$ (x) $\frac{x}{3} + \frac{y}{5}$

2. Find the cube of each of the following by using the formula:

- (i) 18 (ii) 13 (iii) 105 (iv) 10.1 (v) 0.98 (vi) 2.25

3. Find the value of $x^3 + \frac{1}{x^3}$ when:

(i) $x + \frac{1}{x} = 6$ (ii) $x + \frac{1}{x} = -5$

(iii) $x + \frac{1}{x} = \frac{1}{2}$ (iv) $x + \frac{1}{x} = 10.1$

4. Find the value of $x^3 - \frac{1}{x^3}$ when:

(i) $x - \frac{1}{x} = 5$ (ii) $x - \frac{1}{x} = -2$

(iii) $x - \frac{1}{x} = \frac{1}{3}$ (iv) $x - \frac{1}{x} = 9.9$

6.4 SIMULTANEOUS LINEAR EQUATIONS

We know that a linear equation is an equation with degree one. It may be in one, two or more variables.

For example, $2x + 6 = 0$ is a linear equation in one variable x . Similarly $5x + 6y = 9$ is a linear equation in two variables x and y .

Standard form of a linear equation in one variable is: $ax + b = 0$, where a, b are real numbers and $a \neq 0$. For example, $2x + 3 = 0$.

Standard form of a linear equation in two variables is: $ax + by + c = 0$, where a, b, c are real numbers with $a \neq 0$ and $b \neq 0$.

6.4.1 Recognize Simultaneous Linear Equations in one and two variables

Two or more linear equations which occur in one and the same problem are called simultaneous linear equations.

Example 1. $2x + 3 = 5$ and $2x + y = 6$ are two simultaneous linear equations in one and two variables respectively.

Example 2. $6x + y = 8$ and $3x - y = 1$ are two simultaneous linear equations in two variables.

Note: Simultaneous linear equations have only one solution. As in example 1, the solution is $x = 1, y = 4$ or $(x, y) = (1, 4)$. In example 2 the solution is $x = 1, y = 2$ or $(x, y) = (1, 2)$.

6.4.2 Give the concept of formation of a linear equation in two variables.

We know that a linear equation in two variables is in fact an algebraic representation of a problem or statement involving two quantities.

In order to form a linear equation we have to follow following steps.

Step 1. Read the problem or statement carefully and identify the unknown quantities ($x, y, z, t \dots$)

Step 2. Represent the unknown quantities with letters of the English alphabet.

Step 3. Write the equation which satisfies the condition of given problem or statement.

Example 1. Form a linear equation for each of the following statements.

(i) The sum of ages of two students is 30 years.

(ii) Total amount spent for 5 kilogram of apples and 3 kilograms of mangoes is 550 rupees.

(iii) Price of 6 pens is double the price of a book.

Solution: (i) Let the ages of two students in years be x and y . So, the required equation is: $x + y = 30$.

(ii) Let the price of 1kg of apples be x rupees and the price of 1kg of mangoes be y rupees. The required equation will be $5x + 3y = 550$

(iii) Let the price of a pen be x rupees and that of a book be y rupees. So, the required equation will be $6x = 2y$ or $6x - 2y = 0$

Example 2. Form two simultaneous linear equations in two variables for the statement.

The length of a rectangle is greater than its width by 4 cm and perimeter of the rectangle is 192 cm.

Solution: Let length = y and width = x

Then $y - x = 4 \dots$ (i)

Now Perimeter = $2(x + y) = 192$ cm

or $x + y = 96 \dots$ (ii)



6.4.3 Important facts about linear equations know that:

(1) A single linear equation in two unknowns is satisfied by as many pair of values as required.

For example, $x + y = 6$ is a single linear equation in two variables and its solutions are infinite. Some of them are (1, 5), (2, 4), (3, 3), (4, 2) etc.

Verification: We have $x + y = 6$,
 for (1, 5), we get $1 + 5 = 6$ (satisfied) for (2, 4), we get $2 + 4 = 6$ (satisfied)
 for (3, 3), we get $3 + 3 = 6$ (satisfied) for (4, 2), we get $4 + 2 = 6$ (satisfied).

(2) Two simultaneous linear equations in two unknowns have only one solution (i.e., one pair of values)

Either both of these are in two variables or one of them is in one variable.

Note: Solution of two simultaneous equations means the values of the unknowns which satisfy both the equations.

For example, $x + y = 5$ and $x - y = 3$ are two simultaneous linear equations and they have only one solution, i.e., (4, 1).

Verification: We have $x + y = 5$ and $x - y = 3$,
 for (4, 1), we get $4 + 1 = 5$ and $4 - 1 = 3$
 so both the equations are satisfied by (4, 1).

EXERCISE 6.4

1. Identify and separate linear equations in one variable, linear equations in two variables and simultaneous linear equations.

- (i) $2x + 3y = 6$ (ii) $2x - 7 = 0$ (iii) $2x = 6, x + y = 8$
 (iv) $x - y = 7, x + y = 10$ (v) $5y = 8$ (vi) $3y - 7 = 8x$

2. Form a linear equation for each of the following statements:

- (i) Sum of weights of two boys is 50kg.
 (ii) The price of six pens is three times the price of a magazine.
 (iii) If twice of a number is added to 5, the result is 25.
 (iv) If the price of 6kg of mangoes is subtracted from the price of 3kg of guavas, the result is 45 rupees.
 (v) Number of girls in a school is two-third of the number of boys.

3. Guess and verify any three solutions of: (i) $x + y = 10$ (ii) $2x + y = 10$

4. Guess and verify any four solutions of: (i) $x - y = 2$ (ii) $2x + y = 16$

5. Guess and verify the solution of $x - y = 4$ and $x - 2y = 2$

6. Which one of the pair (1, 5) and (6, 0) is not a solution of the simultaneous equations $2x + y = 7$ and $6x - y = 1$?

7. Form a pair of simultaneous linear equations for the following statements:

- (i) The price of a book is 50 rupees more than a notebook and their total price is Rs 115.
 (ii) The sum of two numbers is 27 and difference is 17.
 (iii) Price of a pen is Rs 50 and the price of 2 pens and 5 pencils is Rs 200.
 (iv) The measure of an angle is double that of its complementary angle.

6.5 SOLUTION OF SIMULTANEOUS LINEAR EQUATIONS

There are several methods to solve two simultaneous linear equations, here we will discuss only three methods.

6.5.1 Solve Simultaneous linear equations using

(1) Method of equating the coefficients

We have already learnt the solution of a single linear equation in previous class.

In this method, we make coefficients of a variable equal if they are different and then add or subtract the resultant equations to eliminate that variable as is explained in the following example.

Example 1. Solve $2x - 3y = 3$ and $5x + y = 16$ by using the method of equating the coefficients.

Solution: We have: $2x - 3y = 3$ (i) $5x + y = 16$ (ii)

In order to make co-efficients of x equal. we get:

Multiply eq: (i) by 5, Multiply eq: (ii) by 2

$$10x - 15y = 15 \text{ (iii)}$$

$$10x + 2y = 32 \text{ (iv)}$$

Subtracting eq: (iv) from eq: (iii)

$$10x - 15y = 15$$

$$\pm 10x \pm 2y = \pm 32$$

$$-17y = -17 \text{ (x is eliminated)}$$

$$\Rightarrow 17y = 17 \text{ or } y = \frac{17}{17} = 1.$$

By putting $y = 1$ in eq: (i), we get $2x - 3(1) = 3$

$$\Rightarrow 2x - 3 = 3$$

$$\Rightarrow 2x = 6$$

$$\Rightarrow x = \frac{6}{2} = 3$$

So, $x = 3$ and $y = 1$

Thus, the solution set = $\{(3, 1)\}$.

Note 1: If coefficient of any variable in one of the equations is 1, then multiply only that equation.

Note 2: Try to multiply by smaller numbers.

(2) Method of elimination by substitution

In this method, we find the value of one variable in terms of the other from any one of the equations and use that value in the other equation to eliminate one variable as explained in the following example.

Example 2. Solve: $x + 2y = 8$ and $3x - 4y = -6$ by using the method Of elimination by substitution.

Solution: We have

$$x + 2y = 8 \text{ (i)}$$

$$3x - 4y = -6 \text{ (ii)}$$

From eq: (i), we get: $x = 8 - 2y$ (iii)

By putting $x = 8 - 2y$ in eq: (ii),

$$\text{we get } 3(8 - 2y) - 4y = -6$$

$$\Rightarrow 24 - 6y - 4y = -6$$

$$\Rightarrow 24 - 10y = -6$$

$$\Rightarrow -10y = -6 - 24$$

$$\text{or } -10y = -30$$

$$\text{or } 10y = 30$$

$$\Rightarrow y = \frac{30}{10} = 3.$$

By using $y = 3$ in eq: (iii), we get $x = 8 - 2(3) = 8 - 6 = 2$
 So, $x = 2$ and $y = 3$. Thus, the solution set = $\{(2, 3)\}$

(3) Method of cross-multiplication

In order to explain this method, we take two simultaneous equations in standard form, i.e., $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$. Following are the steps of this method.

Step 1: We write coefficients of variables and constants in the following manner:

$$\begin{array}{ccc} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{array}$$

Step 2: We cross multiply in the following way after excluding first and last column.



Step 3: In order to find values of x and y , we use the following equal ratios.

$$\frac{x}{b_1 c_2 - b_2 c_1} = \frac{y}{c_1 a_2 - c_2 a_1} = \frac{1}{a_1 b_2 - a_2 b_1}$$

Example 3. Solve: $2x + 3y = 4$ and $4x - 5y = -3$ by using method of cross-multiplication.

Solution: We have: $2x + 3y - 4 = 0$ and $4x - 5y + 3 = 0$.
 (Standard form)

Writing the coefficients of the variables and excluding the first and last column or arranging them in the following manner and cross-multiplying, we get



By cross-multiplication method,

$$\frac{x}{(3)(3) - (-5)(-4)} = \frac{y}{(-4)(4) - (3)(2)} = \frac{1}{(2)(-5) - (4)(3)}$$

$$\Rightarrow \frac{x}{9-20} = \frac{y}{-16-6} = \frac{1}{-10-12} \text{ or, } \frac{x}{-11} = \frac{y}{-22} = \frac{1}{-22}$$

Two equations: $\frac{x}{-11} = \frac{1}{-22}$ and $\frac{y}{-22} = \frac{1}{-22}$

$$\Rightarrow -22x = -11 \quad \text{and} \quad y = \frac{22}{22} = 1$$

So, $x = \frac{1}{2}$ and $y = 1$.

Thus, the solution set = $\left\{ \left(\frac{1}{2}, 1 \right) \right\}$ or $\{ (0.5, 1) \}$.

EXERCISE 6.5

1. Solve the following simultaneous equations by using the method of equating the coefficients.

(i) $2x + y = 10$
 $5x - 2y = 7$

(ii) $5x - y = 3$
 $3x + y = 4$

(iii) $3x + 7y = 27$
 $5x + 2y = 16$

(iv) $2x + 3y = 2$
 $4x - 9y = -1$

(v) $2x + y = 8$
 $6x - 2y = -1$

(vi) $2x + 3y = 14$
 $3x + 2y = 5$

2. Solve the following simultaneous equations by using the method of elimination by substitution.

(i) $3x - 5y = 5$
 $2x + y = 12$

(ii) $4x + 3y = 2$
 $5x + 2y = -1$

(iii) $13x - 9y = 1$
 $11x - 12y = 14$

(iv) $7x - 4y = 1$
 $2x - 5y = 8$

(v) $x + y = 1$
 $3x + 6y = 5$

(vi) $3x - 7y = 1$
 $4x + y = 22$

3. Solve the following simultaneous equations by the method of cross-multiplication.

(i) $2x + 5y = 7$
 $5x - y = 4$

(ii) $-2x + 5y = 5$
 $3x - 2y = -13$

(iii) $2x + y = 10$
 $x + 3y = 8$

(iv) $x - 4y = 2$
 $2x - 7y = 5$

(v) $10x + 2y = 8$
 $5x + y = 4$

(vi) $13x - y = 14$
 $2x - 7y = 9$

6.5.2 Solve real life problems involving two simultaneous linear equations in two variables

Let us solve some real life problems involving two simultaneous equations in two variables.

Example 1. Aslam spent Rs 400 to purchase 3 kg of apples and 2 kg of mangoes. Find the price of 1kg of apples if the rate of apples is double the rate of mangoes.

Solution: Let the price of 1 kg of apples be x rupees and the price of 1 kg of mangoes be y rupees.

According to the condition of the problem:

$$3x + 2y = 400 \quad \text{--- (i)}$$

and $x = 2y \quad \text{--- (ii)}$

By putting $x = 2y$ in eq: (i),
we get $3(2y) + 2y = 400$

$$\Rightarrow 6y + 2y = 400$$

$$\Rightarrow 8y = 400$$

$$\Rightarrow y = \frac{400}{8} = 50$$

By putting $y = 50$,
in eq. (ii)

$$\text{we get } x = 2(50) = 100.$$

So, the price of 1kg of apples is 100 rupees.

Example 2. Bisma spent 350 rupees to purchase some pens at the rate of Rs 50 per pen and some copies at the rate of Rs 30 per copy. How many pens did she purchase if the total number of pens and copies was 9?

Solution: Let the number of pens be x and that of copies be y .

According to the condition of the problem:

$$x + y = 9 \quad \text{--- (i)}$$

and $50x + 30y = 350 \quad \text{--- (ii)}$

From eq: (i), $x = 9 - y \quad \text{--- (iii)}$

By putting $x = 9 - y$ in eq: (ii), we get

$$50(9 - y) + 30y = 350$$

$$\Rightarrow 450 - 50y + 30y = 350$$

$$\Rightarrow 450 - 20y = 350$$

$$\text{or } -20y = 350 - 450$$

$$\text{or } -20y = -100$$

$$\text{or } 20y = 100$$

$$\Rightarrow y = \frac{100}{20} = 5.$$

Now by putting $y = 5$ in eq: (iii), we get: $x = 9 - 5 = 4$

So, she purchased 4 pens and 5 copies.

EXERCISE 6.6

1. Aisha spent Rs 300 for purchasing 6 kg potatoes. Find the rate of tomatoes per kg if the rate of tomatoes is thrice the rate of potatoes.
2. Taha purchased bats at the rate of Rs 100 per bat and balls at the rate of Rs 50 per ball. How many balls did he purchase if he spent Rs 600 for eight items?
3. The length of a rectangular playground is double that of its width. If the cost of fencing the whole ground is Rs 9000 at the rate of Rs 30 per metre, find the dimensions of the ground.
4. The number of boys in a school is double that of the girls. What is the number of boys, if the total number of students in the school is 450?
5. The ratio of monthly income of a labourer to its monthly expenditure is 15:14. What is his income, if his monthly saving is Rs 2000?

6.6 ELIMINATION

The method of obtaining a relation from two or more equations, free of a particular variable is called **elimination**.

Note: The relation obtained after elimination is called the eliminant.

**6.6.1 Eliminate a variable from two equations by:
(a) Substitution (b) Application of the formulae.**

There are several methods to eliminate a variable from two equations of two unknowns. However, here we will discuss only two methods.

(i) Elimination by Substitution

In this method, we find the value of a variable (which is to be eliminated) in terms of the other variable from an equation and then that value is put in the other equation to eliminate the required variable.

Example 1. Eliminate x from the following equations:

$$2x + y = 6 \quad \dots\dots (i)$$

$$\text{and } x - 3y = 8 \dots\dots (ii)$$

Solution: We have

$$2x + y = 6 \quad \text{--- (i)}$$

$$x - 3y = 8 \quad \text{--- (ii)}$$

$$\text{From eq: (ii), } x = 8 + 3y$$

$$\begin{array}{l|l}
 \text{By putting } x = 8 + 3y & \Rightarrow 7y = 6 - 16 \\
 \text{in eq: (i), we get} & \Rightarrow 7y + 10 = 0 \\
 2(8 + 3y) + y = 6 & \text{or } 7y = -10 \\
 \Rightarrow 16 + 6y + y = 6 &
 \end{array}$$

Thus, the relation : $7y = -10$ is free of x .

Example 2. Eliminate V_i from the following two equations of motion: $V_f^2 - V_i^2 = 2gS$ and $V_f = V_i + gt$, where V_i and V_f stand for initial and final velocities.

Solution: We have

$$\begin{array}{l|l}
 V_f^2 - V_i^2 = 2gS \rightarrow \text{(i)} & \Rightarrow V_f^2 - (V_f - 2gtV_f + g^2t^2) = 2gS \\
 V_f = V_i + gt \rightarrow \text{(ii)} & \Rightarrow V_f^2 - V_f^2 + 2gtV_f - g^2t^2 = 2gS \\
 \text{From eq: (ii), } V_i = V_f - gt & \Rightarrow g(2V_ft - gt^2) = 2gS \\
 \text{By using } V_i = V_f - gt \text{ in eq: (i),} & \Rightarrow 2V_ft - gt^2 = 2S \\
 \text{we get } V_f^2 - (V_f - gt)^2 = 2gS & \Rightarrow S = \frac{2V_ft}{2} - \frac{gt^2}{2} \\
 & \Rightarrow S = V_ft - \frac{1}{2}gt^2
 \end{array}$$

Thus, the relation $S = V_ft - \frac{1}{2}gt^2$ is free of V_i .

(ii) Elimination by application of formulas

In this method, we apply formulas such as $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$ according to the condition of the problem.

Example 3. Eliminate x from $x + \frac{1}{x} = a$ and $x - \frac{1}{x} = b$.

Solution: We have

$$x + \frac{1}{x} = a \text{ — (i)}$$

$$x - \frac{1}{x} = b \text{ — (ii)}$$

Squaring both sides of eq: (i)

and eq: (ii), we get

$$x^2 + 2 + \frac{1}{x^2} = a^2 \text{ — (iii)}$$

$$\text{and } x^2 - 2 + \frac{1}{x^2} = b^2 \text{ — (iv)}$$

Now subtracting eq: (iv) from eq: (iii), we get

$$x^2 + 2 + \frac{1}{x^2} = a^2$$

$$x^2 - 2 + \frac{1}{x^2} = b^2$$

$$\begin{array}{r}
 - \quad + \quad - \\
 \hline
 4 = a^2 - b^2
 \end{array}$$

Thus, the relation $a^2 - b^2 = 4$ is free of x .

EXERCISE 6.7

1. Eliminate x by substitution method.

(i) $x - y + 1 = 0$
 $3x + y - 3 = 0$

(ii) $x + 5y = 2$
 $6x + 3y = 7$

(iii) $2x + y = 7$
 $x + 3y = 4$

(iv) $x^2 + y = 0$
 $ax + b = 0$

(v) $x - 2y + 1 = 0$
 $3x + y - 3 = 0$

(vi) $4x + 3y + 8 = 0$
 $x + 5y - 2 = 0$

2. Eliminate V_i by substitution method.

(i) $V_f = V_i + at$
 $S = V_i t + \frac{1}{2} at^2$

(ii) $V_f = V_i + gt$
 $S = V_i t - \frac{1}{2} gt^2$

(iii) $V_f^2 - V_i^2 = 2gS$
 $V_f = V_i + gt$

3. Eliminate x by using formula.

(i) $x + \frac{1}{x} = 2a$
 $x - \frac{1}{x} = 2b + 1$

(ii) $x^2 - \frac{1}{x^2} = 2a$
 $x^2 + \frac{1}{x^2} = 3b$

(iii) $\sqrt{x} + \frac{1}{\sqrt{x}} = p$
 $x - \frac{1}{x} = q^2$

(iv) $x^2 + \frac{1}{x^2} = p^2$
 $x^4 + \frac{1}{x^4} = q^4$

(v) $x + \frac{1}{x} = 2y + 1$
 $x - \frac{1}{x} = y - 2$

(vi) $\frac{x^3}{y^3} + \frac{y^3}{x^3} = m$
 $\frac{x^3}{y^3} - \frac{y^3}{x^3} = n$

REVIEW EXERCISE 6

1. Choose the correct answer.

(i) $(x + y)^2 =$ _____.

(a) $x^2 + y^2$ (b) $x^2 + 2xy + y^2$ (c) $x^2 - y^2$ (d) both a and b

(ii) $x^2 - y^2 =$ _____.

(a) $(x + y)^2$ (b) $(x - y)^2$ (c) $x^2 + y^2$ (d) $(x - y)(x + y)$

(iii) $y^3 + 3y^2x + 3yx^2 + x^3 =$ _____.

(a) $(y - x)^3$ (b) $(x - y)^3$ (c) $(x + y)^3$ (d) None of these

(iv) A solution of $x + y + 6 = 0$ is _____.

(a) (1, 2) (b) (-2, 4) (c) (-4, -2) (d) (4, 2)

(v) Solution of the system: $x + y = 5$ and $x - y = 1$ is _____.

- (a) (2, 3) (b) (3, 2) (c) (1, 4) (d) (4, 1)

2. Find the value of: (i) $x^2 + \frac{1}{x^2}$ (ii) $x^4 + \frac{1}{x^4}$ (iii) $x^4 - \frac{1}{x^4}$ (iv) $x^3 + \frac{1}{x^3}$

(v) $x^3 - \frac{1}{x^3}$ (vi) $x^6 - \frac{1}{x^6}$ when $x - \frac{1}{x} = 5$ and $x + \frac{1}{x} = 3$.

3. Evaluate the following using formula:

- (i) $(103)^2$ (ii) $(97)^2$ (iii) $(103) \times (97)$

4. Factorize:

(i) $5x + 10y + 15z$ (ii) $x^2 + 14x + 49$ (iii) $144x^2 - 121y^2$

(iv) $25x^2 - 10xy + y^2$ (v) $a^2 + 4ab + 4b^2 - 9y^2$ (vi) $1 - \frac{1}{x^4}$

5. Find the cube of

(i) $2x + 3y$ (ii) $3x - 4y$ (iii) $x + \frac{1}{x}$ (iv) $x - \frac{1}{x}$

6. Find the cube of:

(i) 29 (ii) 31 by using cube formula.

7. Solve the following system of linear equations by

(a) Using substitution method

(i) $2x - y = 6$, $x + 3y = 10$ (ii) $2x + 5y = 1$, $x - 2y = 4$

(b) Using elimination method

(i) $3x + 2y = 7$, $2x + 3y = 8$ (ii) $3x - 2y = 7$, $5x + y = 29$

(c) Using cross-multiplication method

(i) $2x + y = 5$, $3x - 2y = 4$ (ii) $5x + y = 56$, $x + 18y = 29$.

8. The width of a rectangle is half of its length. Find its area if its perimeter is 24cm.

9. Eliminate x by using formula:

(i) $x - \frac{1}{x} = 2k - 1$; $x + \frac{1}{x} = 5k$ (ii) $x - \frac{1}{x} = m$, $x^2 + \frac{1}{x^2} = n^2$

10. The sum of the digits of a two-digit number is 5. The digits are interchanged and the new number is greater than the original number by 9. Find the number.

(i) By formation of one linear equation in one variable.

(ii) By formation of two simultaneous linear equation in two variables.

SUMMARY

- The process of expressing a polynomial as a product of two or more polynomials is called factorization.
- Some basic formulas:
 - (i) $(a + b)^2 = a^2 + 2ab + b^2$
 - (ii) $(a - b)^2 = a^2 - 2ab + b^2$
 - (iii) $a^2 - b^2 = (a + b)(a - b)$
 - (iv) $(a + b)^3 = a^3 + 3ab(a + b) + b^3 = a^3 + 3a^2b + 3ab^2 + b^3$
 - (v) $(a - b)^3 = a^3 - 3ab(a - b) - b^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- An equation with degree one is called linear equation.
- A linear equation in one variable can generally be written in the form of $ax + b = 0$, for a, b being real numbers with $a \neq 0$.
- Two or more linear equations which occur in one and the same problem are called simultaneous linear equations.
- A linear equation in two variables has infinite solutions.
- The system of two distinct linear equations in two variables may have one or no solution and can be solved by three methods i.e.
- There are several methods to solve simultaneous linear equations viz. elimination method, substitution method and cross-multiplication method.
- The method of obtaining a relation free of a particular variable is called elimination.